

SARANPAUL', Russian FederationWMO: **235270**Lat: **64.253N**Lon: **60.931E**Elev: **95**StdP: **14.65**Time Zone: **5.00 (E05)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-43.6	-38.6	-47.6	0.3	-43.4	-42.4	0.5	-38.4	18.1	5.3	15.7	18.4	0.7	230	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	81.6	65.7	77.8	63.5	74.2	61.7	67.7	76.6	65.8	74.6	63.6	71.8	5.2	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.7	92.0	71.2	62.5	84.7	69.2	60.1	77.8	67.6	32.1	76.6	30.6	74.7	29.0	71.6	73.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.4	14.6	12.3	DB	-45.9	85.9	3.2	2.7	-48.2	87.8	-50.0	89.4	-51.8	90.9	-54.1	92.8	(4)
(5)			WB	-45.7	70.4	3.1	2.4	-47.9	72.2	-49.7	73.6	-51.5	75.0	-53.7	76.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	26.7	-11.5	-1.7	12.8	26.8	41.9	55.4	61.2	54.9	43.5	28.8	9.0	-2.6	
	DBStd	27.09	16.43	15.80	14.41	9.84	9.32	8.53	7.22	6.66	6.87	9.80	16.05	18.99	
	HDD50	9297	1906	1447	1153	696	289	40	6	25	216	658	1230	1631	
	HDD65	14049	2371	1867	1618	1146	716	306	160	321	645	1123	1680	2096	
	CDD50	791	0	0	0	0	38	202	354	176	21	0	0	0	
	CDD65	69	0	0	0	0	0	19	43	7	0	0	0	0	
	CDH74	784	0	0	0	0	30	198	491	64	1	0	0	0	
	CDH80	146	0	0	0	0	2	35	103	6	0	0	0	0	
Wind	WSAvg	4.6	3.3	3.9	5.3	5.8	6.1	5.4	4.4	4.5	4.5	4.2	4.0	3.6	
Precipitation	PrecAvg	19.6	1.0	0.8	0.9	1.3	1.5	2.0	2.9	2.5	2.2	1.8	1.4	1.1	
	PrecMax	27.2	2.3	1.6	2.4	3.6	3.5	5.2	8.0	5.9	5.9	3.5	3.5	3.4	
	PrecMin	14.0	0.4	0.1	0.0	0.2	0.1	0.3	0.7	0.5	0.7	0.1	0.3	0.1	
	PrecStd	3.5	0.5	0.4	0.6	0.7	0.8	1.4	1.7	1.3	1.3	0.8	0.7	0.8	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.1	35.9	43.5	51.1	78.3	84.7	87.1	79.7	69.3	52.2	39.5	35.7	
		MCWB	28.0	32.4	35.2	39.6	59.6	65.5	67.2	66.3	58.0	46.1	36.6	32.0	
	2%	DB	24.2	30.2	39.4	46.0	71.8	79.2	83.7	75.4	62.9	47.7	36.5	32.7	
		MCWB	21.7	27.9	33.8	37.3	55.5	64.4	66.8	63.2	53.9	43.0	33.8	30.5	
	5%	DB	19.3	25.1	36.6	43.4	66.2	75.2	80.0	71.5	58.7	43.8	33.8	28.9	
		MCWB	17.7	22.9	32.0	36.7	53.1	61.8	64.9	61.4	51.9	40.5	30.9	26.8	
	10%	DB	13.4	22.0	33.7	41.0	59.3	70.5	76.3	67.8	55.1	40.9	30.7	24.8	
		MCWB	12.2	19.9	30.3	35.5	48.6	59.2	63.0	59.4	49.5	38.4	28.4	23.0	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.1	32.9	36.7	41.4	61.7	68.8	71.1	68.3	58.6	47.0	36.7	32.5
			MCDB	30.3	35.7	42.6	47.7	76.8	79.1	82.2	76.4	67.9	51.1	38.9	34.9
2%		WB	21.9	27.8	34.2	39.2	57.4	66.2	68.7	65.6	55.2	44.0	34.0	30.6	
		MCDB	24.0	30.1	38.3	44.1	68.8	76.1	78.0	72.1	60.2	47.2	36.5	32.7	
5%		WB	17.8	23.2	32.5	37.6	53.9	63.6	66.6	62.8	52.8	40.9	31.2	27.0	
		MCDB	19.4	24.9	36.3	42.6	64.7	72.3	75.6	68.8	57.9	43.0	33.5	28.6	
10%		WB	12.2	19.9	30.2	35.8	49.7	60.5	65.0	60.5	50.2	38.5	28.3	23.0	
		MCDB	13.4	22.0	33.7	40.6	59.2	68.5	73.9	66.8	54.3	40.6	30.5	24.6	
Mean Daily Temperature Range		5% DB	MDBR	15.4	18.5	20.4	17.8	18.5	18.0	20.0	18.5	14.3	12.0	14.2	15.1
			MCDBR	18.2	17.0	16.7	18.7	30.1	24.1	27.1	24.5	21.7	15.2	14.2	13.6
	MCWBR		17.1	15.1	12.5	12.4	16.4	11.0	11.6	13.2	13.6	10.6	12.5	12.5	
	5% WB	MCDBR	18.2	17.3	15.7	16.0	27.0	21.3	22.9	20.2	19.2	12.3	13.6	13.7	
		MCWBR	17.1	15.7	12.3	11.1	15.9	11.0	10.9	12.2	12.8	8.9	12.2	12.7	
Clear-Sky Solar Irradiance	taub	0.210	0.272	0.313	0.344	0.353	0.366	0.410	0.373	0.330	0.290	0.207	0.211		
	taud	1.924	2.069	2.082	2.127	2.352	2.438	2.311	2.395	2.482	2.359	1.968	1.919		
	Ebn at Noon	106	196	239	260	272	270	252	252	241	198	105	35		
	Edh at Noon	9	20	31	38	33	31	35	29	22	16	8	3		
All-Sky Solar Radiation	RadAvg	50	264	717	1252	1642	1656	1690	1153	611	276	74	18		
	RadStd	3	22	59	115	137	135	134	111	76	31	9	1		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page